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Akihiro Otsuka

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A three-layer model of borrowed size: empirical insights from Japan

Akihiro Otsuka 

Association of International Arts and Sciences, Yokohama City University, Yokohama, Japan

ABSTRACT

This study revisits the concept of borrowed size to explain regional growth beyond that of traditional agglomeration economies. Although scale advantages are typically linked to density, many smaller regions increasingly benefit from relational and institutional ties with metropolitan areas. We propose a three-layer framework comprising scale-related externalities, mediating network connectivity, and institutional and relational foundations. Using data from Japanese prefectures, we demonstrate how improved accessibility, industrial specialisation and national spatial planning enable the diffusion of agglomeration benefits beyond urban cores. Drawing on the Japanese context – characterised by a highly centralised governance structure, pronounced metropolitan concentration and rapid population ageing – we illustrate how national spatial strategy and fiscal transfer mechanisms interact with local industrial specialisations to sustain borrowed-size effects in peripheral regions. These patterns suggest that spatial externalities can operate through non-physical channels, challenging the agglomeration–dispersion dichotomy. Borrowed size is reframed as an active, institutionally mediated process rather than a passive spillover effect, and underscores the role of governance and social capital in shaping inclusive development. Thus, the Japanese case offers transferable insights to other countries undergoing demographic transitions and seeking to balance metropolitan dynamism with regional equity. These findings have practical implications for regional policy in decentralised ageing economies.

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1. Introduction

In recent years, the concentration of economic activity in the Greater Tokyo Area has shown signs of gradual transformation. Notably, the widespread adoption of teleworking in the wake of the COVID-19 pandemic has prompted a reevaluation of the urban-centric lifestyle, opening new perspectives on the residential viability of peripheral regions. However, the structural challenges of population decline and industrial hollowing-out remain acute, calling into question the autonomy and resilience of regional economies. Under such conditions, the conventional explanatory framework based on external economies of agglomeration is becoming increasingly inadequate. The critical issue today is how smaller regions can sustain and develop economic activity by benefiting from scale economies without relying on physical agglomeration.

This study re-examines the concept of ‘borrowed size’ as a theoretical framework to address this question. Borrowed size refers to the phenomenon wherein small-scale regions indirectly benefit from externalities concentrated in large metropolitan areas through various forms of networked or relational proximity. A typical example is a peripheral region that accesses cultural facilities, consumer markets and employment opportunities in a nearby metropolis while maintaining a low-density and cost-effective living environment. Such configurations deviate from the assumptions of the traditional agglomeration theory, which posits that productivity and efficiency depend directly on population density and market size, and instead highlight the importance of networked and relational structures.

The notion of borrowed size also serves as a critical response to theoretical models in the new economic geography literature, such as those developed by Fujita and Tabuchi (1997), who depicted increasing inter-

urban disparities and spatial polarisation. These models suggest that agglomeration economies become self-reinforcing once a threshold is surpassed, justifying economic concentration in large cities. However, recent research findings from Japan indicate that, while economic concentration in major metropolitan areas continues, inter-regional disparities in per capita income have gradually narrowed (Otsuka, 2021, 2024a). This simultaneous occurrence of sustained agglomeration and declining inequality cannot be sufficiently explained by the existing agglomeration or equilibrium dispersion models. In contrast, the borrowed size perspective suggests that small-scale regions can enjoy metropolitan externalities through networks and that this structure may function as a spatial counterbalance to metropolitan congestion and saturation. Against this backdrop, the central research question of this study was as follows:

How can smaller or peripheral urban regions access the benefits of scale economies without undergoing physical agglomeration, and what institutional and relational mechanisms enable such access?

This question lies at the intersection of urban theory, spatial economics and regional policy and challenges conventional assumptions that equate functional capacity with size or density. By addressing this question, this study aims to contribute to a theoretical reorientation that emphasises networked interdependence, institutional embeddedness and nonmaterial linkages to understand regional development.

The concept of borrowed size transcends the binary opposition between spatial concentration and dispersion in urban economics, and can be a mediator when theorising structures in which spatial dispersion and functional integration co-occur. In contemporary regional economies, spatial proximity is not a prerequisite for access to externalities. Instead, non-physical elements, such as institutional connectivity and relational embeddedness, increasingly shape functional scales (Amin & Thrift, 2002; Boschma, 2005; Rodríguez-Pose, 2013). The theoretical framework proposed in this study explicitly incorporates such non-physical conditions of agglomeration and seeks to extend spatial economic models by integrating institutional and relational dimensions.

When discussing borrowed size, it is essential to move beyond the passive reception of spatial spillovers or economic externalities and focus on the institutional and relational foundations that actively mediate such processes. This necessitates an enquiry into the institutional context – policies, regulations and subsidies – and the forms of social capital–inter-firm collaboration, inter-regional networks and human mobility – that facilitate the realisation of borrowed size. This study redefines the concept as a three-layered structure as follows:

- (1) The benefits of scale economies as externalities.
- (2) The mediating function of network connectivity.
- (3) Institutional and relational foundations enabling sustained access to these externalities.

This theoretical framework captures the non-linear and multilayered dynamics of regional growth.

Specifically, this study conceptualises borrowed size as a multilevel phenomenon that operates simultaneously as a *state*, *mechanism* and *process*. First, as a state, borrowed size refers to the *observable outcomes* of functional integration situations in which regions perform beyond what would be expected, given their population or economic scale. Second, as a mechanism, it captures the *causal linkages* through which peripheral or smaller regions access agglomeration benefits through inter-regional connectivity, institutional arrangements and knowledge flows. Third, as a process, it represents the *dynamic evolution* of interactions and adaptations between metropolitan and non-metropolitan regions. In the proposed three-layer framework, these dimensions are hierarchically related; the *state* emerges as the outcome of *mechanisms* that evolve through continuous *processes* of adjustment and institutional mediation. This definitional hierarchy allows for a coherent understanding of borrowed size as both a measurable phenomenon and a theoretical construct.

To underpin this theoretical reconstruction with a descriptive grounding, this study focuses on Japan's economic structure at the prefecture level. The term 'region' refers to prefectural jurisdictions, which are fundamental administrative and statistical units in Japan's spatial policy and economic data systems. This choice was guided by the availability of consistent long-term statistics and the fact that prefectures often act as the primary units of governance for economic and infrastructure planning. However, the theoretical scope of borrowed size in this study extends beyond administrative boundaries to encompass

functional economic areas and inter-regional network systems, particularly in cases where the flows of people, goods and information transcend prefectural lines. Accordingly, prefecture acts as a proxy for a broad set of relational and functional geographies.

This study presents an evidence-based analysis of the trends in real gross value-added (GVA) disparities, spatial proximity and industrial configurations relative to the Greater Tokyo Area, Japan's core region. This study examines the structural conditions under which borrowed size manifests in practice. Rather than seeking a causal inference, the empirical component supplements the theoretical model by illuminating the structural diversity of regional growth that has been overlooked in prior theories.

This study makes three main contributions to the literature. First, it reframes borrowed size as a theory incorporating both institutional thickness and relational connectivity. Second, it grounds this framework in the empirical context of the Japanese regional economy while offering the potential for generalisation to other Asian settings. Third, it proposes an alternative to an agglomeration-centred regional policy by highlighting the possibility of distributed regional growth through network-oriented institutional design and relational integration. In doing so, it seeks to reposition borrowed size as a central theoretical pillar in a new regional growth model that embraces spatial integration and institutional diversity.

The remainder of this paper is organised as follows. Section 2 reviews the existing literature on borrowed size and identifies the theoretical gaps. Section 3 presents the proposed theoretical framework. Section 4 provides empirical evidence, using Japanese prefectural data to assess the applicability of the model. Section 5 explores the policy implications of the framework, and Section 6 concludes the paper and suggests the scope for future research.

2. Rethinking borrowed size: literature review and theoretical gaps

Alonso (1973) introduced the concept of borrowed size in the context of zero urban population growth, referring to the idea that small cities may indirectly benefit from the cultural and economic functions of neighbouring large cities. In the 2010s, Meijers and Burger (2010) re-theorised the notion of embedding it within urban systems theory, particularly with reference to polycentricity and the functional urban region. They redefined borrowed size as a theoretical concept wherein the physical scale of a city does not necessarily align with its functional influence, and where the benefits of agglomeration may be accessed through inter-city networks.

The empirical applications of this framework include a study by Phelps et al. (2001), who quantitatively examined how borrowed size affected the growth and locational dynamics of small firms in the UK. Similarly, Meijers et al. (2016) conducted a multidimensional analysis of European cities to explore the links between functional inter-urban connectivity and economic externalities conferred by borrowed size. These studies demonstrate that borrowed size is not merely a theoretical hypothesis but a concept that can be both observed and measured within actual urban and regional structures.

This perspective resonates with the notion of the 'functional urban region', as advanced by Parr (2007), who emphasises how urban areas may extend their socioeconomic influence through networked relationships, transcending administrative boundaries and physical scale. In this context, borrowed size offers a reconceptualisation of scale that moves beyond static indicators, such as population or GVA, towards a functional reach mediated by inter-urban ties.

Studies such as those by Burger et al. (2015) and Meijers and Burger (2017) further expanded the analytical scope of borrowed size by examining its manifestation through various socioeconomic dimensions, including housing markets, transport networks and industrial structures. These studies validate the conceptual utility of borrowed size and redirect attention toward the latent functional roles of small- and medium-sized cities, long overshadowed in conventional agglomeration theory.

Nevertheless, a substantial portion of the literature continues to frame the mechanisms of borrowed size primarily in terms of spatial proximity and physical distance. The institutional and relational foundations through which borrowed size is mediated remain theoretically underexplored. Questions such as why some regions successfully leverage borrowed size while others do not, and what conditions enable the emergence and persistence of mediating structures, have yet to be systematically addressed.

Furthermore, an understanding of the emergence, evolution and potential dissolution of borrowed size is lacking. Spatial proximity alone does not guarantee access to externalities. However, regions geographically

distant from urban cores may benefit from agglomeration effects if they are underpinned by robust relational capital and institutional complementarity (Camagni et al., 2015). This suggests the need to reconceptualise the mechanisms of spatial externalities through a lens that incorporates institutional and relational structures.

Rodríguez-Pose (2013) highlighted the decisive role of institutional quality and spatial configuration in shaping regional development outcomes, arguing that institutional disparities are the key determinants of the persistence or convergence of territorial inequalities. Rodríguez-Pose and Storper (2006) stressed that the efficacy of institutional change does not depend solely on formal rules but also on their coherence with local social foundations. Similarly, Storper and Scott (2016) identified the fragmented treatment of institutions, cultures and networks as a critical challenge for contemporary urban theory, reinforcing the theoretical value of framing borrowed size within an institutional mediation structure.

The function of borrowed size should be examined concerning its potential antithesis—negative externalities, particularly agglomeration shadows. Krugman (1991) and Fujita et al. (1999), within the context of new economic geography, modelled the centripetal forces of agglomeration that draw resources away from peripheral regions, leading to economic hollowing-out. Tabuchi (1998) synthesised Alonso's land-use theory using Krugman's agglomeration framework to quantitatively map the spatial distribution of such effects. Brakman et al. (2004) empirically identified the long-term imprint of agglomeration shadows on post-war urban growth using the natural experiment on wartime bombing in Germany.

Borrowed size and agglomeration shadow are the 'light and shadow' of spatial externalities. In territories lacking a sufficient network structure or institutional mediation, agglomeration benefits are unlikely to diffuse, borrowed-size effects are marginalised, and agglomeration shadow becomes dominant. These structural bifurcations are essential for any reconsideration of spatial configuration and redesign of regional policy.

In Japan, where a clear urban hierarchy persists and decades of infrastructure development and institutional engineering have shaped territorial dynamics, it is reasonable to assume that borrowed size is governed less by physical distance and more by institutional proximity and relational connectedness. This view is consistent with the theoretical frameworks of Amin and Thrift (2002) and Storper (1997); they emphasised the importance of cultural and institutional ties in shaping inter-urban relationships.

Against this background, this study seeks to reconstruct a theoretical framework of borrowed size. Specifically, it conceptualises its mechanisms through a threefold structure:

- (1) The benefits of scale economies as externalities.
- (2) Mediating function of network connectivity.
- (3) institutional and relational foundations provide sustained access to these externalities.

First, scale-related externalities, such as productivity gains, labour market thickening and knowledge spillovers, have traditionally been attributed to population size and spatial concentration. However, this study suggests that such effects may be transmitted across broader spatial extents via networks without necessitating physical contiguity (Boschma, 2005; Camagni et al., 2015).

Second, network-mediated connectivity encompasses spatial distance, institutional and social links that facilitate the flow of information, labour and capital. Functional connectivity among regions depends on physical infrastructure, such as transport or information and communications technology (ICT), and on trust-based relationships and institutional agreements (Bathelt et al., 2004; Storper & Scott, 2016).

Third, institutional and relational foundations refer to policy frameworks, inter-regional coordination mechanisms and support schemes that structurally sustain network operations. Such institutional arrangements transform borrowed size from a transitory effect into a constitutive element of regional resilience (Aoki, 2001; Rodríguez-Pose, 2013).

By redefining borrowed size as a three-layer model, this study offers a theoretical framework that explains the coexistence of persistent agglomeration and narrowing spatial inequality – an empirical paradox that conventional spatial economics has struggled to resolve.

Although Camagni et al. (2015) and Boschma (2005) discussed the theoretical foundations of borrowed size, the present framework advances this discussion in two important directions. First, unlike Camagni et al.'s focus on agglomeration-related interdependencies within functional urban areas, this study builds a multilayered framework incorporating functional, symbolic and institutional dimensions across a wider territorial

scale. Second, while Boschma emphasises proximity and relatedness in knowledge flows, the present framework integrates these concepts into explicit territorial governance mechanisms, particularly relevant in non-metropolitan contexts, where institutional borrowing is more salient than economic clustering.

3. A three-layer model of borrowed size

This section presents a theoretical reconstruction of the borrowed size concept by clarifying its multilayered structure. While previous studies have predominantly understood borrowed size in terms of functional inter-urban linkages, often emphasising spatial proximity and the visible structure of networks, they have paid less attention to the underlying mechanisms by which the economic benefits of scale are transferred to smaller regions. To elucidate how such benefits reach less populated areas, borrowed size must be conceptualised as a three-layered structure comprising the source of externalities, transmission mechanisms and the institutional foundations that support them.

Figure 1 shows the conceptual framework of borrowed size as a three-layered structure integrating scale-related externalities, network connectivity and institutional-relational foundations. This framework illustrates how small or peripheral regions can access and sustain the benefits of metropolitan externalities through mediated and embedded mechanisms.

These three layers do not function in isolation. Instead, they are mutually interdependent and co-evolve within territorial configurations. This co-evolutionary process manifests borrowed size as an operative force within real-world regional dynamics.

Figure 2 presents the hierarchical logic of the borrowed size to further clarify the conceptual architecture of the proposed framework. The *state* denotes the measurable outcomes of regional integration, where peripheral or smaller regions perform beyond their expected size. The *mechanism* represents the causal and institutional linkages enabling these outcomes, and the *process* captures the dynamic interactions through which such mechanisms evolve. Figure 1 shows the empirical configuration of the three layers, and Figure 2 focuses on their theoretical interdependencies.

In this hierarchy, *processes* denote the ongoing dynamics through which inter-regional interactions, institutional adjustments and learning mechanisms evolve. These *processes* give rise to *mechanisms* – network-mediated, relational and institutional linkages that enable the peripheral regions to access the benefits

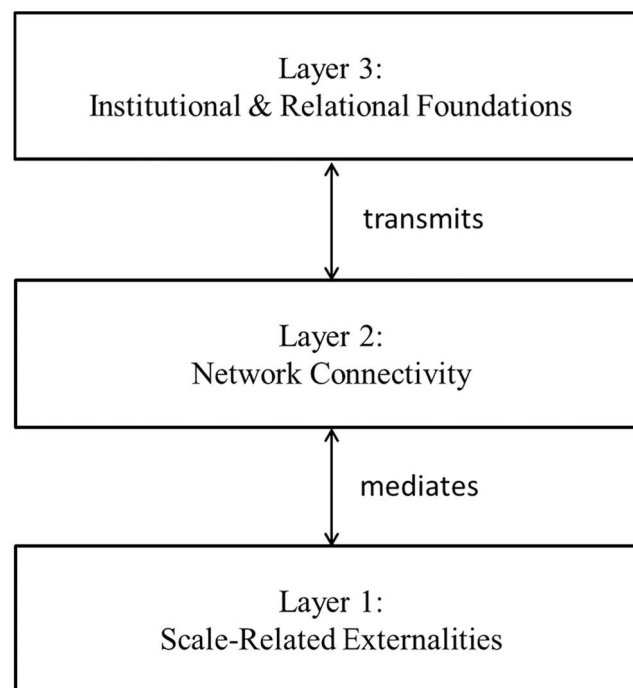


Figure 1. Conceptual diagram of the three-layer borrowed size framework.

Source: Author's elaboration.

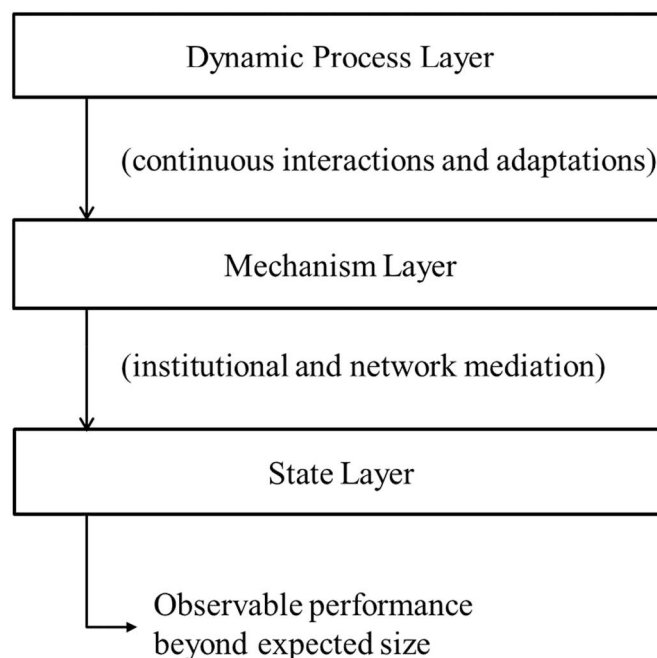


Figure 2. Conceptual hierarchy of borrowed size.

Source: Author's elaboration.

of agglomeration. The *state* reflects the observable outcome of these mechanisms, regional performance beyond that expected from the population or economic scale. Importantly, this relationship is not unidirectional; feedback from observed *states* often reshapes the underlying *processes*, creating an iterative system of adaptation and transformation. This conceptual hierarchy helps reconcile different interpretations of borrowed size found in the literature – whether as a condition, a causal mechanism, or an evolving process – by integrating them into a single analytical structure.

3.1. Scale economies as externalities

The first element of borrowed size is the benefit of scale, that is, the external economies arising from agglomeration. These include thick labour markets, diversified industrial clusters, knowledge and technological spillovers, and the capacity to accommodate heterogeneous consumer preferences. Metropolitan regions are particularly adept at internalising such externalities, functioning as engines of economic efficiency and innovation.

Traditionally, these externalities have been understood to be an outcome of spatial concentration. However, in today's context, geographical distance no longer represents a principal constraint. Thus, the benefits of scale are not confined to physically agglomerated areas. Instead, they can be functionally reconstituted through extensive networks and institutional linkages. This perspective necessitates a redefinition of the concept of scale, repositioning borrowed size from a notion grounded in a substantive scale to one based on a relational scale (Camagni et al., 2015).

Here, the 'substantive scale' refers to the economic scale determined by physical attributes such as population size and spatial extent. In contrast, the term 'relational scale' denotes a functional economic space characterised by nonmaterial forms of connectivity, including inter-urban networks and institutional ties. Mediating network structures are essential for reconfiguring economies of scale in a manner that does not rely on physical proximity.

3.2. Mediating structures of network connectivity

The second element of borrowed size concerns the mediating structures through which externalities are transmitted from metropolitan centres to peripheral regions, namely, the connectivity of inter-regional

networks. It involves spatial proximity and structural linkages that facilitate the movement of people, goods and information. Transport infrastructure, ICTs, supply chains and inter-institutional collaborations, particularly among universities and research institutes, constitute essential conditions for the realisation of borrowed size (Phelps et al., 2001).

The extent to which such networks are institutionally embedded and underpinned by relational trust. For instance, in interdependent industrial clusters, where labour mobility and interfirm transactions are intensive, informal conventions and mutual trust, rather than formal contracts, are conduits for the transmission of externalities.

Bathelt et al. (2004) distinguished between ‘local buzz’ – the informal circulation of knowledge within regions – and ‘global pipelines’ – structured connections to actors beyond the region. These dual mechanisms operate complementarily to generate innovation and foster growth. Similarly, the mediating structures of borrowed size rely on this dual architecture, in which the density of local networks and the quality of external linkages together constitute the critical determinants of regional development.

This second layer, centred on physical and functional connectivity, is a critical channel through which peripheral regions can tap into the core’s market size and concentration-driven externalities that constitute the first layer. However, network connectivity should be understood not only in its technical or physical sense, but also as a socially constructed and institutionally negotiated framework through which borrowed size can materialise. Although these mediating structures can transmit agglomeration benefits from the core to the periphery, they can also serve as conduits for the reverse flow of economic gains back into the core. Without mechanisms for financial and resource circulation back to the periphery, these benefits may leak, limiting local multiplier effects, a phenomenon often referred to as the agglomeration shadow or ‘leaky bucket’ problem in regional development (New Economics Foundation, 2002). Therefore, the second layer alone is insufficient to ensure sustainable regional growth. Third, an institutional layer, encompassing fiscal transfers, revenue-sharing schemes and other systemic arrangements, is necessary to retain and reinvest these flows in peripheral economies. Overcoming such negative externalities requires institutional complementarity as a prescription, marking a conceptual shift beyond infrastructural provisions toward the qualitative dimensions of network architecture as a core principle in contemporary spatial economic theory.

Accordingly, network connectivity must be understood not only in terms of its technical or physical properties, but also as a socially constructed and institutionally negotiated framework through which borrowed size can materialise. This perspective calls for a conceptual shift beyond infrastructural provisions, emphasising the qualitative dimensions of network architecture as a central tenet of contemporary spatial economic theory.

3.3. Institutional and relational foundations

The third element of borrowed size is the institutional and relational underpinnings that ensure the stability and functionality of inter-regional networks. Borrowed size must not be conceived merely as a temporary appropriation of externalities but rather as a sustainable and reproducible regional configuration. Non-market factors such as institutional frameworks (comprising policy agreements, fiscal transfers and regulatory flexibility) and social embeddedness (including social capital, cultural affinity and normative trust) play a decisive role (Amin & Thrift, 2002).

A rigorous theoretical understanding of these institutional and relational conditions draws on institutional economics literature. First, as North (1990) emphasised, institutions are path-dependent constructs formed through the accumulation of historical decisions, and institutional change occurs within historically contingent frameworks. Second, Rodrik (2007) argued for complementarity between institutions and policies, suggesting that institutional arrangements conducive to economic development must exhibit *institutional coherence*; that is, they must align with the socioeconomic context of a given region or nation.

Finally, Rodríguez-Pose (2013) argued that it is not the abstract quality or content of institutions per se that determines regional development outcomes, but rather their *institutional fit* – how well they are embedded and functionally integrated within the specific context of the local society. This insight cautions against the wholesale transplantation of institutional models and calls for a context-sensitive institutional design.

Aoki (2001) conceptualised institutions as systems of mutually complementary components. The coherence of expectations and the sharing of rules across these components guide institutional stability and

evolutionary change. His notion of *institutional complementarities* offers a robust conceptual foundation for understanding the ‘interoperability’ among the institutional arrangements necessary to sustain borrowed size.

Together, these perspectives affirm that borrowed size is not merely a by-product of spatial networks but a phenomenon actively mediated and sustained through institutional configurations. The durability of networks and their capacity to convey externalities fundamentally depend on the design and complementarity of the institutional infrastructure upon which they rest. Without such institutional arrangements, the benefits transmitted through the first (scale economies) and second (network connectivity) layers are prone to leak back to the core, diminishing their impact on peripheral economies. This dynamic is captured by the leaky bucket analogy of regional development, wherein the income generated through integration is rapidly lost through outflows, undermining the local multiplier effects. This phenomenon is also linked to the concept of an agglomeration shadow, such as the absorption of regional resources into core regions.

In the Japanese context, the Local Allocation Tax historically sought to correct such disparities through fiscal transfers. Local allocation tax is a transfer of funds from the central government to local governments. This is the revenue that local governments can use freely, and it plays an important role in local government finances. In addition to the local allocation tax, the *Furusato Nozei* programme (hometown tax donation), introduced in 2008, provides a more voluntary, demand-driven channel for resource circulation, enabling urban residents to redirect part of their tax payments to rural municipalities (Ministry of Internal Affairs and Communications, *n.d.*). These schemes exemplify the institutional devices that can counterbalance the outflows caused by agglomeration shadows and reinforce the sustainability of borrowed-size effects.

In this regard, institutions should not be treated as passive background conditions but as *enabling mechanisms* – structural devices that render the circulation of externalities feasible and effective. Scott and Storper (2003) advanced this view by underscoring regional development’s critical interplay among institutions, cultures and economic systems. In the face of intensifying global competition, they contended that institutional embeddedness is a key determinant of regional resilience. This reinforces the proposition that the institutional-relational foundations of borrowed size are indispensable for both its emergence and long-term viability.

Peripheral actors typically pursue normative objectives such as balanced development, service accessibility, demographic retention and resilience. Realising these aims depends on concrete institutional capacities – fiscal space and transfer design, administrative coordination capability and network-brokerage competences – through which cooperative arrangements can be sustained and conflicts (e.g., inter-jurisdictional tax competition) are managed over time.

3.4. Conceptual model: multi-scalar borrowed size mechanism

By integrating the three constituent elements outlined above, the borrowed size mechanism can be conceptualised as a multilayered structure encompassing the following components:

- (A) Source (scale economies): Concentration of economic and intellectual externalities in metropolitan cores.
- (B) Mediation structure (network): The transmission of these externalities to surrounding regions through physical, institutional and relational connectivity.
- (C) Reception base (institutions and relations): Institutional, cultural and trust-based foundations are necessary for the stable absorption and utilisation of these externalities.

The integration of these three layers is grounded in the theory of institutional complementarity (Aoki, 2001; North, 1990), which posits that the productivity of one component is enhanced when combined with other mutually reinforcing components. In the context of regional growth, physical agglomeration economies (Layer 1) and network connectivity (Layer 2) can generate short-term gains. However, without governance and relational capacity (Layer 3), such gains are prone to dissipation due to failures in coordination, fiscal leakage and inter-jurisdictional competition. This complementarity implies that the absence of any layer reduces the marginal returns of the others, thereby establishing a theoretical rationale for treating them as an inseparable framework.

These three layers do not function in isolation; rather, they operate in an interdependent and co-evolutionary manner such that the borrowed size emerges only when all the components align and reinforce one another. In the absence of institutional support for the mediating structure, scale benefits may fail to reach peripheral regions. Worse still, as Brakman et al. (2004) demonstrated, such asymmetries may lead to the dominance of negative externalities, manifested as an agglomeration shadow. In such cases, the same inter-regional interdependencies that could generate positive borrowed-size effects instead manifest as agglomeration shadows: negative net externalities flowing toward the core at the expense of the periphery (Otsuka, 2024b, 2025). Borrowed size and agglomeration shadow are therefore not independent phenomena but represent two possible outcomes of a single mechanism, differentiated by the balance of inflows and outflows of externalities. Strong institutional and relational capacities, such as equitable fiscal transfer schemes, reinvestment of locally generated revenues and governance arrangements fostering mutual trust, shift this balance toward sustained positive effects. Conversely, weak or asymmetric institutional frameworks increase the likelihood of dominating the agglomeration shadow effects.

Ultimately, borrowed size should be redefined not merely as a concept within urban network theory, but as an integrative theoretical device that links institutions with spatial configurations. Assessing how spatial and institutional linkages within the Greater Tokyo Area, along with industrial complementarities, shape the dynamics of regional growth will offer important insights into the empirical validity and relevance of the model to policy.

4. Regional structures in Japan: empirical patterns and descriptive evidence

This section presents a descriptive analysis of the inter-regional economic structures in Japan to generate empirical insights into the three-layer framework of borrowed size. Particular attention is given to trends in economic disparity at the prefectural level, as measured by the GVA, along with spatial distance from the Greater Tokyo Area and patterns of industrial specialisation. These observations aim to shed light on the seemingly paradoxical coexistence of sustained agglomeration and the narrowing of regional disparities, an outcome that may be interpreted through the lens of the borrowed size mechanism. It is important to note that the purpose of this descriptive enquiry is not to establish causal relationships per se but rather to complement the preceding theoretical discussion by identifying the structural characteristics and qualitative trends that signal the applicability and relevance of the conceptual framework.

4.1. GVA convergence and divergence: a paradoxical trend

Table 1 illustrates the regional distribution of the GVA (real values) and population across Japan as of 2019. The data reveal a marked concentration of economic activity in the Greater Tokyo Area, accounting for 33.53% of the national GVA, which is 4.64 percentage points higher than its population share of 28.89%. This indicates that Tokyo's economic productivity per capita was significantly higher than the national

Table 1. Gross value added and population shares by region (2019).

	GVA (real, million yen)	Percentage share (%)	Population (person)	Percentage share (%)
Hokkaido	19,998,297	3.47	5,253,681	4.18
Tohoku	43,896,424	7.62	10,934,957	8.69
Northern Kanto	36,220,222	6.28	7,547,924	6.00
Greater Tokyo Area	193,281,798	33.53	36,332,943	28.89
Chubu	83,872,547	14.55	16,956,915	13.48
Hokuriku	13,360,482	2.32	2,948,367	2.34
Kansai	88,043,096	15.28	20,477,181	16.28
Chugoku	30,717,200	5.33	7,319,688	5.82
Shikoku	14,688,595	2.55	3,730,052	2.97
Kyushu	47,772,674	8.29	12,815,889	10.19
Okinawa	4,524,201	0.78	1,455,353	1.16

Notes: The regional classifications were as follows: Hokkaido (Hokkaido); Tohoku (Aomori, Iwate, Miyagi, Akita, Yamagata, Fukushima and Niigata); Northern Kanto (Ibaraki, Tochigi, Gunma and Yamanashi); Greater Tokyo Area (Saitama, Chiba, Tokyo and Kanagawa); Hokuriku (Toyama, Ishikawa and Fukui); Chubu (Nagano, Gifu, Shizuoka, Aichi and Mie); Kansai (Shiga, Kyoto, Osaka, Hyogo, Nara and Wakayama); Chugoku (Tottori, Shimane, Okayama, Hiroshima and Yamaguchi); Shikoku (Tokushima, Kagawa, Ehime and Kochi); Kyushu (Fukuoka, Saga, Nagasaki, Kumamoto, Oita, Miyazaki and Kagoshima); and Okinawa (Okinawa).

Source: Cabinet Office, Government of Japan, Prefectural-level Economic Database.

average. Kansai (15.28%) and Chubu (14.55%) followed in both GVA and population share. This spatial pattern reflects the concentration of scale economies in metropolitan cores.

By contrast, regions outside these three major metropolitan areas (Hokkaido, Hokuriku, Shikoku and Okinawa) account for less than 10% of both the GVA and population, highlighting the substantial economic and demographic disparities between the core and peripheral regions.

Table 2 presents the top five prefectures in terms of GVA and population. Tokyo ranks highest in both categories, with an especially dominant position in GVA at 19.75% of the national total. Aichi and Osaka follow at around 7%, but the gap with Tokyo exceeds 10 percentage points. In terms of population, Tokyo (10.89%) is followed by Kanagawa (7.27%) and Osaka (6.96%), with less disparity than in GVA. This reinforces that Tokyo's economic concentration exceeds what its population share alone would suggest. Combined, the top five prefectures account for 44.12% of national GVA and 36.88% of the population – a 7.24 percentage point difference, indicating stronger centralisation of economic activity than of population. Compared to 2014, the shares of these leading prefectures have risen in both metrics, pointing to an intensification of regional concentration in recent years.

Turning to inter-regional disparity trends, Figure 3 depicts changes in the coefficient of variation (CV) for per capita GVA in Japan. The CV declined sharply between 1990 and 1995, suggesting rapid convergence of regional economic performance. Although a temporary uptick occurred in 2005, the CV resumed its decline thereafter, reaching by 2019 a level significantly below that of 1990. This indicates a clear long-term trend toward narrowing disparities in per capita GVA among regions.

Further insights are provided in Figure 4, which plots per capita GVA levels in 1990 against subsequent growth rates up to 2019. A pronounced negative correlation emerges: regions with relatively lower GVA in 1990 recorded higher growth rates over the following decades. This pattern, indicative of β -convergence, reflects economic catch-up processes in less developed regions.

Taken together, these trends suggest that although the Greater Tokyo Area (Tokyo, Kanagawa, Chiba and Saitama) has maintained high levels of economic activity, several non-metropolitan regions have recorded notable improvements since the 2010s. The declining CV confirms a gradual convergence in regional economic performance, even alongside sustained agglomeration at the core.

This dual trend – the simultaneous persistence of agglomeration and the mitigation of spatial inequality – contradicts the self-reinforcement predicted by conventional spatial economic theory. To resolve this apparent paradox, it is necessary to examine how the first layer (scale economies) interacts with the second layer (mediating networks) and the third layer (institutional and relational foundations) of the borrowed size mechanism.

4.2. Spatial proximity and asymmetric connectivity

Japan's high-speed transport infrastructure has developed into a radial network centred on the Greater Tokyo Area. Transport systems, including highways, Shinkansen (bullet trains) and air routes, have been systematically developed, with Tokyo as the national hub. Since the 1990s, this network has expanded rapidly, substantially reducing the travel times between Tokyo and many regional cities. These reductions in temporal distance represent not only physical improvements, but also a functional expansion of the metropolitan economic sphere, thereby enhancing the potential for borrowed-size effects in peripheral regions.

Table 2. Top five prefectures by GVA and population (2019).

Prefecture	GVA share (%)	Prefecture	Population share (%)
Tokyo	19.75	Tokyo	10.89
Aichi	7.17	Kanagawa	7.27
Osaka	7.06	Osaka	6.96
Kanagawa	6.06	Aichi	5.97
Saitama	4.08	Saitama	5.78
Top5 sum (2019)	44.12	Top5 sum (2019)	36.88
Top5 sum (2014)	42.79	Top5 sum (2014)	35.50

Note: The table includes comparative data for 2014.

Source: Cabinet Office, Government of Japan, Prefectural-level Economic Database.

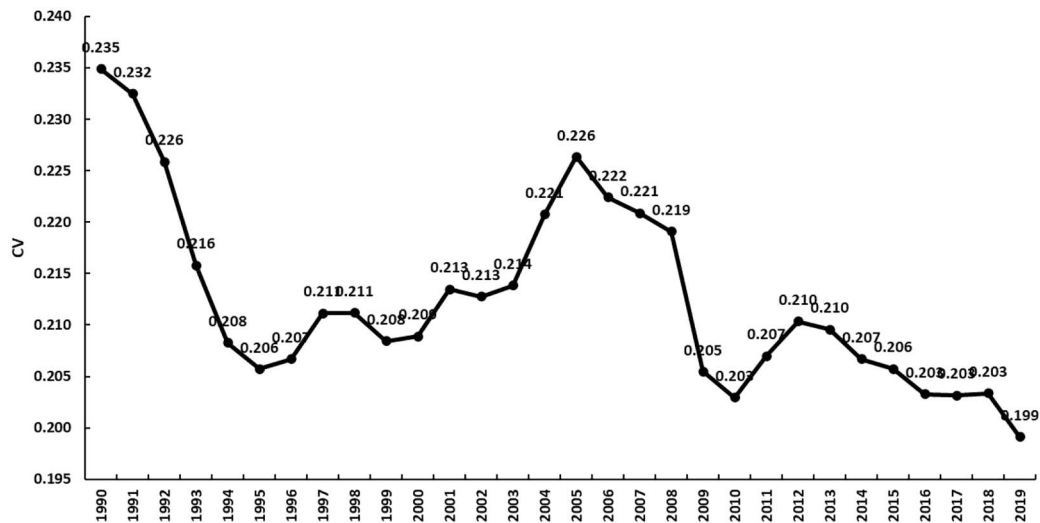


Figure 3. Changes in the coefficient of variation of per capita GVA.

Source: Cabinet Office, Government of Japan, Prefectural-level Economic Database.

Figure 5 presents the relationship between each region's travel time to Tokyo and its labour productivity growth rate between 1990 and 2019. As the scatterplot suggests, regions farther from Tokyo experienced relatively higher productivity growth. This counterintuitive pattern suggests that improvements in connectivity disproportionately benefit remote areas, enabling them to capture previously inaccessible agglomeration externalities.

Figure 6 illustrates the correlation between the degree of reduction in travel time to Tokyo and labour productivity growth over the same period. A statistically significant positive correlation (correlation coefficient: 0.35) is observed, supporting the proposition that improvements in temporal proximity have contributed to productivity gains. Notably, this effect is most pronounced in more peripheral prefectures such as Tokushima, Yamaguchi, Shimane, Aomori and Iwate. This finding aligns with the second layer of the

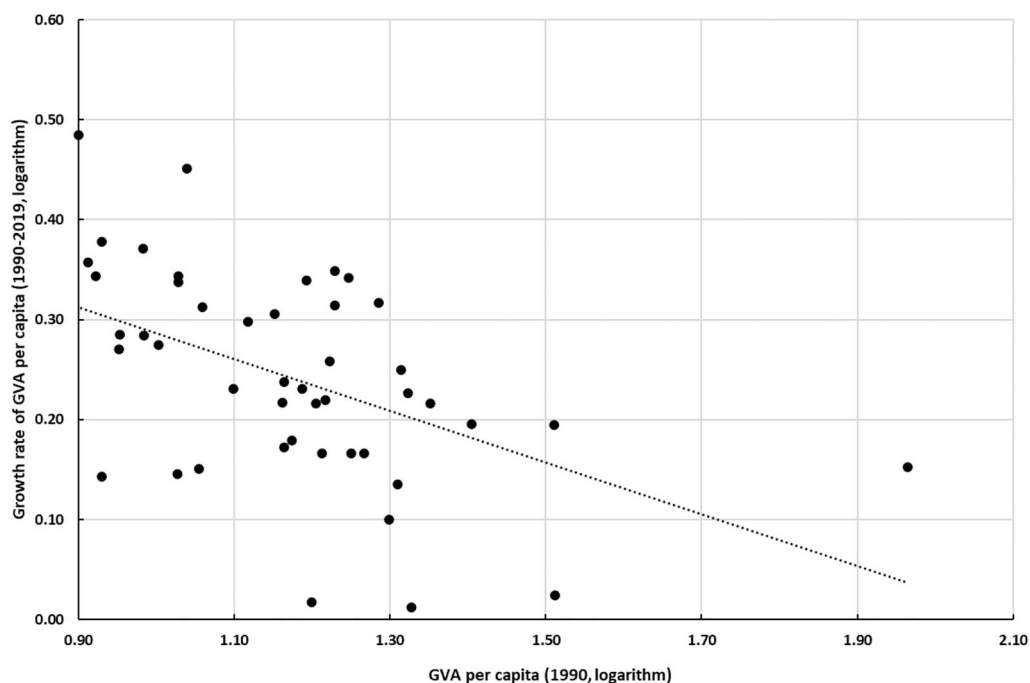


Figure 4. Relationship between initial per capita GVA and growth rate (1990–2019).

Source: Cabinet Office, Government of Japan, Prefectural-level Economic Database.

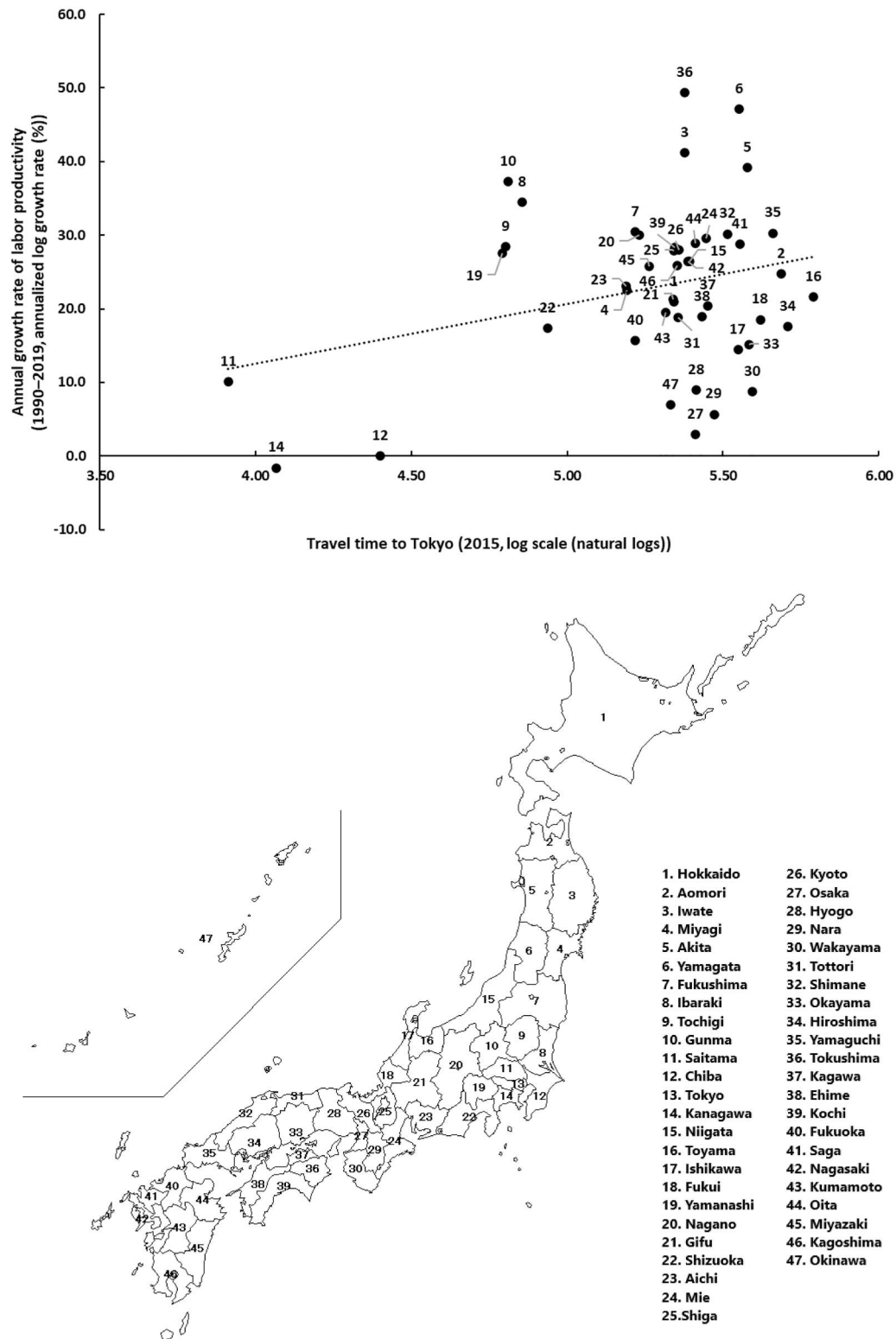


Figure 5. Time distance to Tokyo and labour productivity growth by region. Note: Prefecture number and name.

Source: (a) Cabinet Office, Government of Japan, Prefectural-level Economic Database. (b) Ministry of Land, Infrastructure, Transport, and Tourism (MLIT), Government of Japan, National Integrated Transport Analysis System (NITAS).

conceptual framework – mediating network structures – whereby reductions in cost to access expand the reach of metropolitan spillovers beyond their immediate hinterlands.

These findings suggest that many regions demonstrating convergence in productivity levels have benefited from enhanced functional connectivity to Tokyo via high-quality transport infrastructure, such

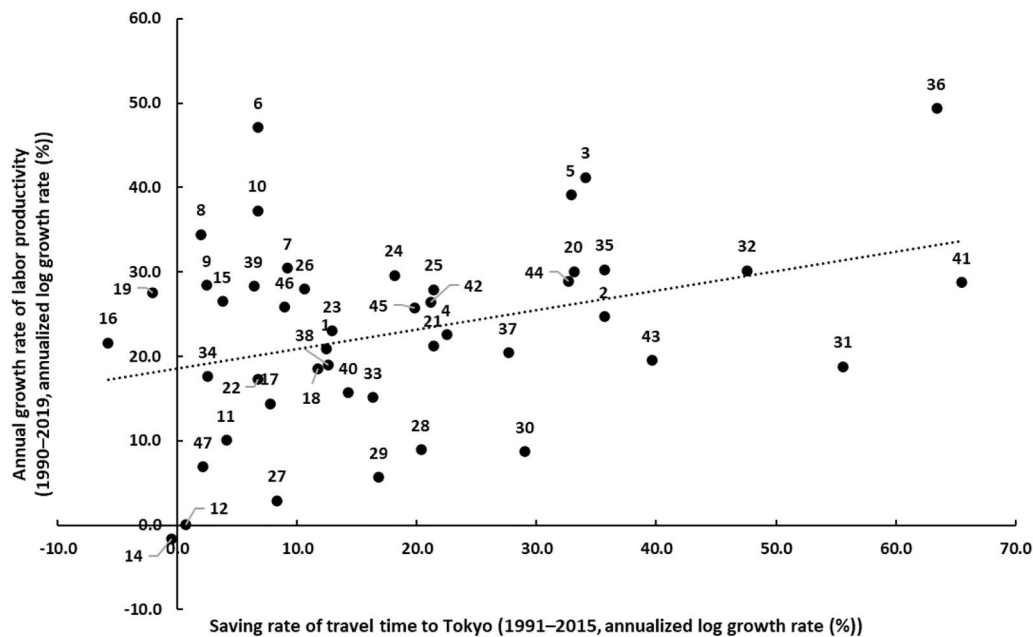


Figure 6. Reduction in time distance and growth in labour productivity. Note: The correspondence between prefecture numbers and prefecture names is shown in the map in Figure 5.

Source: (a) Cabinet Office, Government of Japan, Prefectural-level Economic Database. (b) Ministry of Land, Infrastructure, Transport, and Tourism (MLIT), Government of Japan, National Integrated Transport Analysis System (NITAS).

as highways and high-speed rail services. However, these benefits cannot be explained by measures of distance alone. They are likely underpinned by institutional arrangements, inter-organisational agreements and social trust – what might be termed ‘relational infrastructure’ – facilitating the effective absorption of transmitted externalities.

This also implies that spatial spillovers do not diffuse uniformly across space, but are filtered through asymmetric network geometries. Temporal proximity may be a necessary but insufficient condition and the translation of proximity into tangible economic gains depends on the density, redundancy and resilience of the underlying network as well as the presence of policy and governance mechanisms that sustain these linkages over time.

Thus, reductions in travel time, when coupled with well-embedded institutional and relational frameworks, may enable physically remote regions to benefit from scale economies in ways not predicted by conventional distance-decay models. The following section examines the role of industrial structures and institutional conditions in supporting spillover absorption.

4.3. Industrial specialisation and relational complementarities

An examination of the industrial composition of Japan’s regions reveals that many peripheral areas experiencing narrowing economic disparities have undergone industrial specialisation, particularly in manufacturing. This specialisation acts as an ‘absorptive interface’ through which external scale economies can be internalised, aligning with the reception base described in the conceptual model.

Figure 7 illustrates the strong positive correlation between the degree of specialisation in manufacturing and the rate of labour productivity growth. For instance, prefectures such as Shiga, Tokushima, Gunma and Ibaraki exhibit high levels of manufacturing agglomeration and above-average productivity growth despite not being part of the major metropolitan areas. This pattern suggests that peripheral regions can achieve productivity catch-up when sectoral advantages allow them to effectively channel and amplify metropolitan spillovers.

Such trends can be attributed to the technological characteristics and market structures of individual sectors. Manufacturing, being capital-intensive and well-aligned with ICT investment and process innovation, tends to facilitate productivity enhancement even in regions with declining populations. In contrast, service

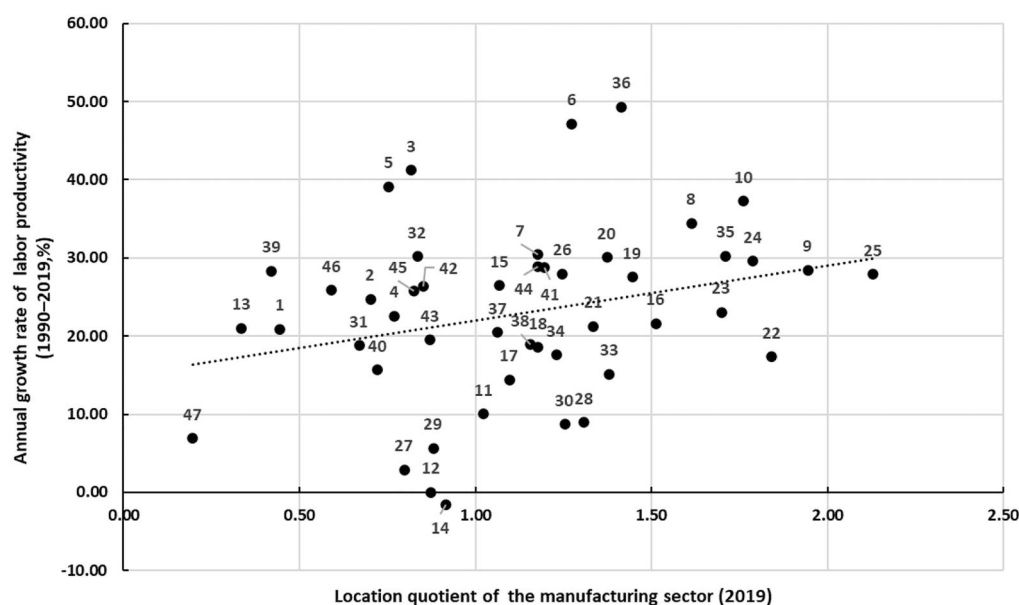


Figure 7. Manufacturing specialisation and productivity growth. Note: The correspondence between prefecture numbers and prefecture names is shown in the map in Figure 5.

Source: Cabinet Office, Government of Japan, Prefectural-level Economic Database.

industries are generally labour-intensive and heavily reliant on local demand, making them more susceptible to population-related constraints on productivity. In other words, manufacturing-intensive regions possess a structural resilience that enables them to convert externalities from metropolitan networks into sustained productivity gains.

Importantly, this pattern of regional manufacturing specialisation complements the industrial structures of major metropolitan areas. Rather than a unidirectional dependency on large cities, a strategic division of labour across regions emerges. Such complementarities enhance production efficiency and deepen inter-regional linkages, reinforcing the mediation layer of the borrowed size framework. For example, small and medium-sized enterprises in rural areas may be embedded within urban supply chains, or regional production hubs may specialise in outputs destined for urban markets.

Camagni et al.'s (2015) analysis of the conditions under which second-ranked cities can benefit from agglomeration economies is particularly instructive. They demonstrated that such cities could harness the advantages of scale by maintaining functional connectivity in large urban regions and institutional foundations. This finding supports the theoretical framework of borrowed size, particularly the dual emphasis on relational complementarity and institutional mediation.

Capello (2009) argued that traditional spatial economic theories tend to underestimate the role of institutional frameworks and network connectivity in explaining regional growth. From this perspective, the three-layer structure of the borrowed size model supplements the existing growth models with institutional depth.

Crucially, these relationships are not confined to market transactions but also rely on institutional and organisational linkages. Collaborative arrangements between academia, industry and the government in research and development (R&D), inter-regional economic agreements and sustained support from regional financial institutions all serve as the foundational components of institutional mediation (Bristow & Healy, 2015). These mechanisms stabilise production networks and provide governance structures that secure the continuous circulation of externalities, thereby operationalising the third layer of the borrowed size mechanism.

In Japan, the institutional and relational foundations for inter-regional collaboration are embedded in the National Spatial Strategy (NSS), which is a statutory nationwide plan formulated under the National Land Planning Act. As a long-term comprehensive spatial plan for the use, development and conservation of land at the national level, the NSS provides a framework that specifies spatial policies, infrastructure

priorities and governance arrangements across prefectures and metropolitan areas, thereby functioning as a formalised platform for realising relational externalities beyond individual administrative boundaries.

Since the first comprehensive national development plan of 1962, national spatial planning in Japan has been periodically revised to address emerging challenges, evolving from infrastructure-led growth to network-oriented, resilience-focused strategies. The current NSS (Ministry of Land, Infrastructure, Transport and Tourism, [n.d.](#)), with a ten-year planning horizon, defines its core concept as the creation of a 'country that promotes interaction', where 'interaction' refers to the inter-regional movement of people, goods, capital and information arising from cooperative relationships. These interactions, mediated by inter-regional networks, are considered essential for sustainable regional development. In the context of population decline, the NSS emphasises maintaining the regional market size through networking, which requires preserving the individuality of each region as an economic entity. By refining their distinct local resources, regions can generate mutually beneficial exchanges that enhance national productivity and efficiency. A notable example is the planned Chuo Shinkansen, a 438 km Maglev line linking Tokyo, Nagoya and Osaka. Once operational (Tokyo–Nagoya in 2037, Nagoya–Osaka in 2045), travel times between these major metropolitan areas will be reduced to approximately one hour, allowing face-to-face collaboration, joint research and innovation across an integrated 'super-mega-region'. This illustrates how institutional frameworks (Layer 3) and physical connectivity (Layer 2) can act in concert to strengthen the overall spatial economy.

In addition to the NSS, Japan's institutional foundations for inter-regional collaboration are reinforced by two long-standing statutory mechanisms that directly shape the absorptive capacity of peripheral regions.

First, intergovernmental fiscal transfer systems such as the Local Allocation Tax function as formalised redistribution mechanisms that equalise fiscal capacity across municipalities. [Figure 8](#) shows the change in the CV of the per capita GVA before and after fiscal transfers. The figure clearly shows a downward shift after fiscal transfers, indicating a narrowing of regional disparities. By securing a stable revenue base for less-affluent jurisdictions, these transfers enable the continued provision of public services and strategic investments in local infrastructure. This institutional support sustains the economic viability of peripheral areas and enhances their ability to participate in inter-regional production networks, thereby increasing the likelihood that external-scale economies from metropolitan areas will be effectively absorbed and retained.

Second, the NSS framework facilitates large-scale social capital investments, including transport, communication and logistics infrastructure. As documented in the post-war NSS, infrastructure development under these plans has improved market accessibility, reduced transaction costs and supported the formation of regional industrial hubs ([Otsuka, 2021](#), pp. 30–33). Such investments illustrate how Layer 3 (institutional

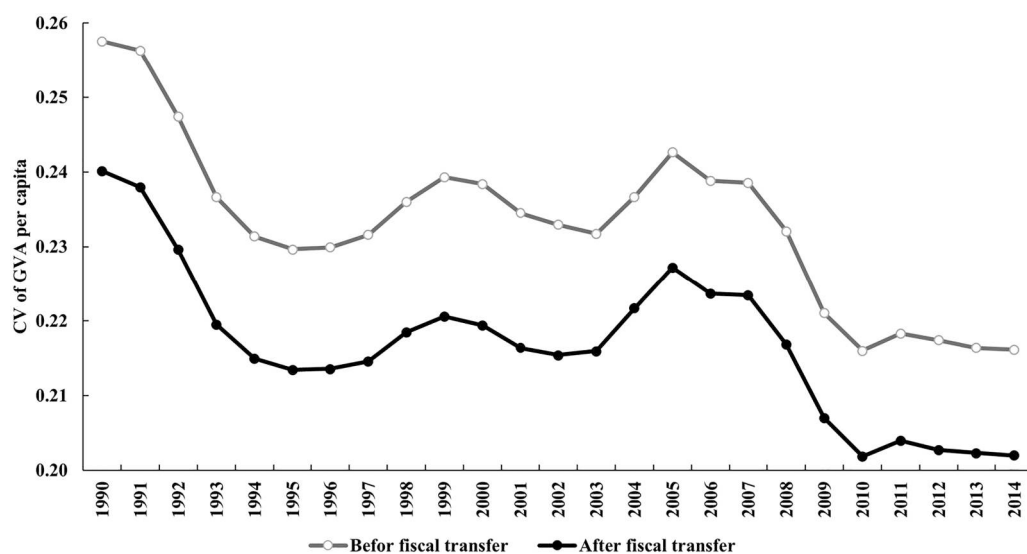


Figure 8. Changes in the coefficient of variation before and after fiscal transfer.

Source: Calculations based on *the Annual Report on Prefectural Accounts* (Cabinet Office), *Basic Resident Registers* (Ministry of Internal Affairs and Communications) and *the Nikkei Economic Electronic Databank System*.

frameworks) operates in synergy with Layer 2 (physical connectivity), creating the structural conditions necessary for dynamic inter-regional interaction.

Together, these fiscal and planning mechanisms constitute tangible institutional evidence for Layer 3 in the borrowed size framework. Strengthening the absorptive capacity of the peripheral regions through financial resources and planned connectivity facilitates the transformation of spatial proximity and network linkages into sustained productivity gains.

In summary, regional industrial specialisation and institutional structures operate as key absorptive capacities that enable the effects of borrowed size to materialise. The interplay between sectoral advantages and embedded institutional arrangements determines whether spatial proximity and connectivity translate into tangible economic growth.

4.4. Linking empirical patterns to theoretical insights

The preceding descriptive analyses do not merely catalogue empirical facts but serve to substantiate the theoretical coherence of the three-layer borrowed size framework. Specifically:

- The gradual convergence of GVA across regions suggests that, although the source of external economies remains concentrated in core areas such as Tokyo, a structural mechanism for their redistribution via transport networks and industrial linkages is empirically observable.
- The reduction in the time distance and improvements in functional connectivity underscore the central role of network connectivity as a mediating structure in shaping the growth potential of peripheral regions.
- The presence of complementary industrial linkages and institutional support mechanisms demonstrates that institutional and relational foundations transform the concept of borrowed size from a passive receipt of externalities to a structurally sustainable process, especially when such institutional arrangements are combined with physical and relational connectivity to form a mutually reinforcing growth system.

Together, these empirical patterns reinforce the explanatory power and policy relevance of the borrowed size framework for capturing the complexities of regional development in contemporary Japan. They highlight that scale-related benefits are not simply ‘spilled over’ but are actively filtered, shaped and amplified through spatial networks and institutional capacities. They also provide a conceptual bridge for rethinking growth theory in spatial economics, particularly in terms of how scale-related benefits are distributed across space through non-linear multi-scalar mechanisms.

In the next section, we build on this integration of theory and empirical observations to explore the implications of borrowed size for regional policy design and theoretical advancement.

5. Theoretical and policy implications

5.1. Rethinking spatial scale: from physical to relational constructs

The three-layer model of borrowed size offers both theoretical and practical implications for spatial economics. Its core contribution lies in reconceptualising the spatial scale, traditionally understood in purely physical terms, as a relational construct shaped by institutional and network-mediated channels.

Classical models in new economic geography (Krugman, 1991; Fujita et al., 1999) explain growth through the self-reinforcing effects of agglomeration: dense populations and large markets generate centripetal forces, often increasing interurban disparities. This powerful logic assumes that external economies diffuse primarily through geographical proximity.

However, contemporary Japan shows that sustained agglomeration can coexist with narrowing regional disparities. This pattern suggests that scale economies are redistributed through institutional arrangements and inter-regional networks, not solely through physical closeness. This corresponds to the first layer of the model, scale-related externalities, illustrated by the long-term convergence in per-capita GVA in Japan despite persistent metropolitan concentration, as discussed in Section 4.1. Therefore, scale should be conceptualised as relational, with its benefits depending on the presence of mediating structures that link the core and peripheral regions.

Incorporating institutional mediation into the spatial economic theory expands its explanatory scope, allowing for a more integrated understanding of how agglomeration effects diffuse under specific structural conditions. This reconceptualisation moves beyond the geography of distance toward the geography of institutional connectivity.

5.2. Integrating institutions into spatial economic theory

Redefining spatial scale as relational also requires shifting the analytical focus from whether institutions exist to how they function within specific regional contexts. Rodríguez-Pose (2013) emphasised that institutional fit – the alignment of institutions with local conditions – is more important than generic notions of ‘good institutions’. This directly resonates with the borrowed size framework, which assumes that the diffusion of scale economies depends on institutions tailored to the territories they serve.

Rodríguez-Pose and Storper (2006) contended that institutional effectiveness stems not only from formal design but also from embeddedness in informal rules, such as social trust, relational networks and cultural norms. This aligns with Aoki’s (2001) concept of institutional complementarities, where mutually reinforcing institutions create coherent expectations and collective actions. In the borrowed-size framework, these complementarities are critical for sustaining spillovers over time.

Storper and Scott (2016) called for a theory that integrates institutions, networks and space. Borrowed size can act as a nexus, linking physical and relational scales to institutional depth. This corresponds to the third layer of the model – institutional and relational foundations – and in the Japanese context, examples include the NSS and the fiscal transfer system noted in Section 4.3. By embedding institutional analysis into spatial economic theory, borrowed size extends existing growth models (Capello & Nijkamp, 2009), providing a more comprehensive account of the structural mechanisms that shape regional development in contemporary economies.

5.3. Conceptualising agglomeration shadow and borrowed size as dual outcomes

Borrowed size and agglomeration shadow can be understood as dual interrelated outcomes of inter-urban networks. Agglomeration processes often generate centripetal forces that boost growth in urban cores but suppress development in peripheral areas – what Fujita et al. (1999) and Krugman (1991) describe as the ‘shadow effect’.

This study proposes that institutional and relational networks can counterbalance this shadow. When such mediating structures are present, the benefits of scale, normally concentrated in the core, can be redistributed to peripheral regions, thus enabling borrowed size. In this sense, borrowed size is not merely an incidental by-product of agglomeration but an institutional counterweight to its negative spillovers. This duality connects the second layer (network-mediated connectivity) and the third layer (institutional foundations), both of which must operate together to offset shadow effects.

Framing these phenomena as structurally intertwined, yet conditionally distinct, clarifies that spatial inequalities are not inevitable outcomes of market forces. Instead, they depend on the presence or absence of mediating institutional arrangements. This perspective implies that policy must intentionally design scale through coordination, network planning and institutional bridging to transform disparities into functional complementarities. By adopting this dual-outcome lens, spatial policy can move beyond reactive redistribution toward proactive integration, supporting both spatial justice and the resilience of inter-urban systems.

5.4. Four strategic pillars of policy design

The three-layer borrowed size model demonstrates that regions can capture scale benefits without relying on physical agglomeration, thus offering a framework for reimagining regional development beyond core-centred growth. Instead of concentrating resources on metropolitan hubs, policy can leverage institutional reinforcement and relational connectivity to foster distributed and resilient growth. Each of the four strategic pillars is mapped to at least one layer of the model to ensure a direct link between theory and policy design. Four strategic pillars emerge for policy design:

5.4.1. Ensure institutional interoperability (Layer 3)

Harmonise regulations, improve interjurisdictional governance and strengthen administrative coordination. These measures reduce the friction between municipalities and provide a structural foundation for functional integration.

5.4.2. Cultivate relational capital (Layers 2 and 3)

Promote human mobility, support inter-regional academic–industrial networks and institutionalise trust-based partnerships. Relational capital deepens the institutional complementarity and enhances the adaptability of networked regions.

5.4.3. Redesign functional scales (Layers 1 and 2)

Move beyond administrative boundaries toward policy frameworks aligned with actual economic and living spheres. This recalibration allows institutional configurations to match territorial realities.

5.4.4. Enhance mediating structures (Layer 2)

Strengthen intermediary organisations, regional hubs and public–private partnerships that enable innovation and coordinate cross-regional cooperation. These entities serve as operational enablers for borrowed size.

In operationalising these pillars, Boschma's (2005) five proximities – geographical, organisational, institutional, social and cognitive – offer an analytical lens for assembling diverse forms of proximity into coherent regional strategies.

This approach contrasts with traditional agglomeration-focused policy, which entrenches dependency on peripheral areas. By prioritising institutional, relational and functional spatial logic, a policy based on the borrowed size framework can transform spatial disparities into complementarities, advancing both equity and efficiency in regional systems. As shown in Sections 4.2 and 4.3, this approach is particularly relevant in Japan, where high-speed transport connectivity and industrial specialisation provide a foundation for networked growth. Although grounded in the Japanese context, the conceptual and policy insights derived from the borrowed size framework hold broader applicability for analysing distributed growth and network-mediated externalities in other national and regional settings, particularly in countries facing similar demographic and spatial restructuring.

6. Conclusion and future research agenda

This study proposes a theoretical reconceptualisation of borrowed size as a three-layered construct encompassing scale-related externalities, mediating network structures and institutional-relational foundations. This multi-scalar, non-linear framework offers an integrated interpretation of the seemingly paradoxical coexistence of sustained agglomeration and declining inter-regional disparities observed in Japan, thereby suggesting a new direction for regional policy paradigms.

By shifting the analytical focus from agglomeration intensity to inter-municipal and institutional interdependence, this study extends the borrowed size theory beyond the interpretations of Camagni et al. (2015) and Boschma (2005). Descriptive evidence from Japan underscores how peripheral regions can benefit from institutional coordination, shared governance platforms and relational capital even in the absence of physical proximity or functional dominance. These findings contribute to the growing recognition that regional resilience and growth are increasingly contingent on nonmaterial forms of connectivity.

The policy implications of this study should be interpreted in light of a *process-oriented* understanding of borrowed size. Regional resilience is not a static condition but an outcome of continuous institutional mediation and network adaptation. Therefore, effective regional policy should focus less on reproducing metropolitan agglomeration externally and more on facilitating dynamic processes such as institutional cooperation, governance coordination and knowledge circulation that enable regions to sustain functional integration over time. Empirical illustrations from Japanese prefectures support this process-oriented interpretation of borrowed size and demonstrate how institutional coordination enables functional integration beyond physical agglomeration.

However, this study has three primary limitations. First, its high level of theoretical abstraction raises questions regarding balancing conceptual clarity and empirical precision. Second, the operationalisation and measurement of the institutional and relational components of borrowed size remain methodologically underdeveloped, requiring further advancement in quantitative approaches. Third, the scope is confined to prefecture-level cross-sectional indicators, which may not fully capture microscale interactions or the temporal dynamics of institutional change. This study does not aim to provide causal inference or formal validation but rather to construct a conceptual foundation and illustrate its contextual relevance in the Japanese setting.

Future studies should address these limitations. First, dynamic modelling using evolutionary spatial network simulations and longitudinal datasets is warranted to capture the formation, deepening and dissolution of borrowed size. Second, a multi-scalar analysis spanning municipalities, metropolitan regions and wider functional areas may provide insight into the scope and scale dependence of the mechanism. Third, comparative institutional studies across diverse national and regional contexts would clarify the universality or context-specificity of borrowed size, particularly in light of the socially and historically contingent nature of institutions. Such cross-national perspectives are especially relevant for countries undergoing demographic transition, metropolitan concentration and spatial restructuring, where network-mediated spillovers may function to counterbalance spatial polarisation. Fourth, new relational indicators such as mobility trajectories, supply chain linkages and university–industry collaborations are needed to refine the empirical analyses.

This three-layered model contributes to the reconfiguration of contemporary urban and regional theory by situating borrowed size as a mediating structure at the intersection of institutions, culture and space. As Rodríguez-Pose and Storper (2006) contended, institutions are not merely formal rules but social constructs embedded in norms and trust networks that shape regional development. This perspective reinforces the theoretical relevance of conceptualising borrowed size as a regionally constructed and institutionally mediated growth mechanism. Although this study focuses on Japan, the proposed three-layer framework may be applicable to other countries with metropolitan concentration and emerging inter-regional linkages. By acknowledging its methodological boundaries and outlining the pathways for empirical expansion, this framework provides a robust platform for cumulative research across spatial and institutional contexts.

In conclusion, the borrowed size offers a novel theoretical foundation for understanding growth mechanisms beyond physical agglomeration. Harnessing institutional and relational networks provides a new perspective for designing strategies for regional resilience and inclusive development. Future research should continue to refine this framework as both a theoretical cornerstone and practical model for spatial policy grounded in institutional resilience.

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Research ethics and consent

This study did not involve human participants or identifiable personal data. Therefore, ethical approval and informed consent were not required.

Data availability statement

The data supporting the findings of this study are derived from publicly available sources provided by the Cabinet Office of Japan and the Statistics Bureau of Japan. Processed datasets and analytical codes are available from the author upon reasonable request.

ORCID

Akihiro Otsuka  <http://orcid.org/0000-0001-7852-4590>

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